

Geometría diferencial y aplicaciones

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Escuela Técnica Superior de Ingeniería

Nombre y apellidos:

Teoría: Diferencial covariante.

1.- Sea $(M, J, g, \omega, \nabla)$ una variedad Hermítica, con su métrica Riemanniana Hermítica g , con su estructura compleja J , con su forma de Kähler ω , y con su conexión de Levi-Civita ∇ . Demostrar que:

$$2g((\nabla_X J)Y, Z) = d\omega(X, JY, JZ) - d\omega(X, Y, Z)$$

para cualesquiera campos $\{X, Y, Z\} \subset \mathfrak{X}(M)$.

2.- Sea la región M de $\mathbb{T}^2$ en el afín euclídeo, o sea $M \hookrightarrow \mathbb{T}^2 \hookrightarrow \mathbb{A}^3$, parametrizada por $\vec{x}$, con $R > 1$:

$$\begin{array}{ccc} (0, 2\pi) \times (0, 2\pi) & \xrightarrow{\vec{x}} & \mathbb{R}^3 \\ (\theta, \varphi) & \mapsto & \vec{x}(\theta, \varphi) = ((\cos \varphi + R) \cos \theta, (\cos \varphi + R) \sin \theta, \sin \varphi) \end{array}$$

Sean las 2-formas de curvatura $\Omega_i^j \in \Omega^2(M)$, las cuales están definidas tal que: $R(X, Y) \frac{\partial}{\partial x^i} = \sum_{j=1}^2 \Omega_i^j(X, Y) \frac{\partial}{\partial x^j}$ con cualesquiera par de campos $\{X, Y\} \subset \mathfrak{X}(M)$, $x^1 = \theta$, $x^2 = \varphi$. Demostrar, sin usar el lema de Poincaré, que la clase de cohomología de de Rham $[\Omega_1^1 + \Omega_2^2] \in H_{dR}^2(M)$ es nula.

① Desarrollo un lado de la igualdad:

$$d\omega(X, \nabla Y, \nabla Z) - d\omega(X, Y, Z) \stackrel{?}{=} \text{Definición sintética de la diferencial exterior.}$$

$$= X \overset{①}{\omega(\nabla Y, \nabla Z)} + (\nabla Y) \overset{②}{\omega(\nabla Z, X)} + (\nabla Z) \overset{③}{\omega(X, \nabla Y)} - \overset{④}{\omega([X, \nabla Y], \nabla Z)} + \overset{⑤}{\omega([X, \nabla Z], \nabla Y)} - \overset{⑥}{\omega(\nabla Y, \nabla Z), X} - \overset{⑦}{X \omega(Y, Z)} - \overset{⑧}{Y \omega(Z, X)} - \overset{⑨}{Z \omega(X, Y)} + \overset{⑩}{\omega([X, Y], Z)} - \overset{⑪}{\omega([X, Z], Y)} + \overset{⑫}{\omega(Y, Z), X}$$

Desarrolla el otro lado:

$$2g(\nabla_X \nabla Y, Z) \stackrel{?}{=} \text{Igualdad (ejercicio de clase)}$$

$$= 2g(\nabla_X \nabla Y, Z) - 2g(\nabla_X \nabla Y, Z) \stackrel{?}{=} \text{Teorema fundamental de la geometría.}$$

$$= X g(\nabla Y, Z) + (\nabla Y) g(X, Z) - Z g(X, \nabla Y) - g(X, [\nabla Y, Z]) - g(\nabla Y, [X, Z]) + g(Z, [\nabla Y, X]) - 2g(\nabla_X \nabla Y, Z) = *$$

$$g(\nabla_X \nabla Y, Z) \stackrel{?}{=} g(\nabla^2(\nabla_X Y), \nabla Z) = g(\nabla_X Y, \nabla Z)$$

↑
g métrica hermitica

$$* = X \overset{⑦}{g(\nabla Y, Z)} + (\nabla Y) \overset{②}{g(X, Z)} - Z \overset{⑧}{g(X, \nabla Y)} - \overset{①}{g(X, [\nabla Y, Z])} - \overset{⑩}{g(\nabla Y, [X, Z])} + \overset{④}{g(Z, [\nabla Y, X])} + X \overset{⑨}{g(Y, \nabla Z)} + Y \overset{③}{g(\nabla Z, X)} - (\nabla Z) \overset{⑤}{g(X, Y)} + \overset{⑥}{g(X, [Y, \nabla Z])} - \overset{⑫}{g(Y, [X, \nabla Z])} - \overset{⑪}{g(\nabla Z, [Y, X])}$$

↑
Teorema Fundamental de la geometría

Basta ver que los 12 sumandos de los dos desarrollos coinciden:

$$① \mathcal{L} \omega(\mathcal{J}Y, \mathcal{J}Z) = \mathcal{L} g(\mathcal{J}Y, \mathcal{J}^2 Z) = \mathcal{L} g(Y, \mathcal{J}Z)$$

$$② (\mathcal{J}Y) \omega(\mathcal{J}Z, X) = (\mathcal{J}Y) g(\mathcal{J}Z, \mathcal{J}X) = (\mathcal{J}Y) g(Z, X)$$

$$③ (\mathcal{J}Z) \omega(X, \mathcal{J}Y) = (\mathcal{J}Z) g(X, \mathcal{J}^2 Y) = -(\mathcal{J}Z) g(X, Y)$$

$$④ -\omega([X, \mathcal{J}Y], \mathcal{J}Z) = -g([X, \mathcal{J}Y], \mathcal{J}^2 Z) = g([X, \mathcal{J}Y], Z) = -g(Z, [\mathcal{J}Y, X])$$

$$⑤ \omega([X, \mathcal{J}Z], \mathcal{J}Y) = g([X, \mathcal{J}Z], \mathcal{J}^2 Y) = -g(Y, [X, \mathcal{J}Z])$$

$$⑥ -\omega([\mathcal{J}Y, \mathcal{J}Z], X) = -g([\mathcal{J}Y, \mathcal{J}Z], \mathcal{J}X) \quad (**)$$

$$⑦ -X \omega(Y, Z) = -X g(Y, \mathcal{J}Z) = -X g(\mathcal{J}Y, \mathcal{J}^2 Z) = X g(\mathcal{J}Y, Z)$$

$$⑧ -Y \omega(Z, X) = -Y g(Z, \mathcal{J}X) = -Y g(\mathcal{J}Z, \mathcal{J}^2 X) = Y g(\mathcal{J}Z, X)$$

$$⑨ -Z \omega(X, Y) = -Z g(X, \mathcal{J}Y)$$

$$⑩ \omega([X, Y], Z) = g([X, Y], \mathcal{J}Z) = -g(\mathcal{J}Z, [Y, X])$$

$$⑪ -\omega([X, Z], Y) = -g([X, Z], \mathcal{J}Y)$$

$$⑫ \omega([Y, Z], X) = g([Y, Z], \mathcal{J}X) \quad (**)$$

(**) Falta ver que son iguales ⑥+⑫ y ⑩+⑪

$$-g([\mathcal{J}Y, \mathcal{J}Z], \mathcal{J}X) + g([Y, Z], \mathcal{J}X) = g(-[\mathcal{J}Y, \mathcal{J}Z] + [Y, Z], \mathcal{J}X) \quad ⑥+⑫$$

$$-g(X, [\mathcal{J}Y, Z]) - g(X, [Y, \mathcal{J}Z]) = g(-X, [\mathcal{J}Y, Z] + [Y, \mathcal{J}Z]) \quad ⑩+⑪$$

$$⑩+⑪ = g(\mathcal{J}(-X), \mathcal{J}([\mathcal{J}Y, Z] + [Y, \mathcal{J}Z])) = g(-\mathcal{J}X, \mathcal{J}([\mathcal{J}Y, Z] + \mathcal{J}[Y, \mathcal{J}Z])) =$$

$$= g(\mathcal{J}X, -\mathcal{J}[\mathcal{J}Y, Z] - \mathcal{J}[Y, \mathcal{J}Z]) = g(-\mathcal{J}[\mathcal{J}Y, Z] - \mathcal{J}[Y, \mathcal{J}Z], \mathcal{J}X) =$$

$$= g(-[\mathcal{J}Y, \mathcal{J}Z] + [Y, Z], \mathcal{J}X) = ⑥+⑫$$

↑

$N_{\mathcal{J}} = 0 \equiv$ Campo de Nijenhuis
de $\mathcal{J}$ es nulo.

q.e.d.

$$② \quad R(\underline{x}, y) \frac{\partial}{\partial x^i} = \sum_j \Omega_j^i(\underline{x}, y) \frac{\partial}{\partial x^j}$$

Se trata de demostrar que $\exists \omega \in \Omega^1(M)$ tal que $d\omega = \Omega_1 + \Omega_2$
 Lo haremos por cálculo directo:

1.º Busco la expresión de Ω_1^1 y Ω_2^2 en coordenadas

$$R(\underline{x}, y) \frac{\partial}{\partial x^1} = \sum_j \Omega_j^1(\underline{x}, y) \frac{\partial}{\partial x^j} \Rightarrow dx^1 \left(R(\underline{x}, y) \frac{\partial}{\partial x^1} \right) = \Omega_1^1(\underline{x}, y) =$$

$$= \sum_{ij} (\Omega_1^1)_{ij} dx^i \otimes dx^j(\underline{x}, y) \Rightarrow$$

$$\Rightarrow (\Omega_1^1)_{ij} = dx^1 \left(R \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) \frac{\partial}{\partial x^1} \right) = dx^1 \left(\sum_k R_{ij1}^k \frac{\partial}{\partial x^k} \right) = R_{ij1}^1 \Rightarrow$$

$$\Rightarrow \Omega_1^1 = \sum_{ij} R_{ij1}^1 dx^i \otimes dx^j$$

$$\text{Igualmente } R(\underline{x}, y) \frac{\partial}{\partial x^2} = \sum_j \Omega_j^2(\underline{x}, y) \frac{\partial}{\partial x^j} \Rightarrow dx^2 \left(R(\underline{x}, y) \frac{\partial}{\partial x^2} \right) = \Omega_2^2(\underline{x}, y)$$

$$= \sum_{ij} (\Omega_2^2)_{ij} dx^i \otimes dx^j(\underline{x}, y) \Rightarrow$$

$$\Rightarrow (\Omega_2^2)_{ij} = dx^2 \left(R \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) \frac{\partial}{\partial x^2} \right) = dx^2 \left(\sum_k R_{ij2}^k \frac{\partial}{\partial x^k} \right) = R_{ij2}^2 \Rightarrow$$

$$\Rightarrow \Omega_2^2 = \sum_{ij} R_{ij2}^2 dx^i \otimes dx^j$$

2.º Calculo los coeficientes R_{ij1}^1, R_{ij2}^2

$$R_{ij1}^1 = \frac{\partial \Gamma_{j1}^1}{\partial x^i} - \frac{\partial \Gamma_{i1}^1}{\partial x^j} + \sum_\alpha (\Gamma_{j1}^\alpha \Gamma_{i\alpha}^1 - \Gamma_{i1}^\alpha \Gamma_{j\alpha}^1)$$

$$R_{ij2}^2 = \frac{\partial \Gamma_{j2}^2}{\partial x^i} - \frac{\partial \Gamma_{i2}^2}{\partial x^j} + \sum_\alpha (\Gamma_{j2}^\alpha \Gamma_{i\alpha}^2 - \Gamma_{i2}^\alpha \Gamma_{j\alpha}^2)$$

2º a) Busco las métricas inducidas g y g^{-1} de M con $x^1 = \theta$, $x^2 = \varphi$.

$$\left\langle \frac{\partial \vec{x}}{\partial \theta}, \frac{\partial \vec{x}}{\partial \theta} \right\rangle = E = g_{11} = (\cos \varphi + R)^2, \left\langle \frac{\partial \vec{x}}{\partial \theta}, \frac{\partial \vec{x}}{\partial \varphi} \right\rangle = F = g_{12} = g_{21} = 0$$

$$\left\langle \frac{\partial \vec{x}}{\partial \varphi}, \frac{\partial \vec{x}}{\partial \varphi} \right\rangle = G = g_{22} = 1 \Rightarrow g^{11} = \frac{1}{(\cos \varphi + R)^2}, g^{22} = 1, g^{12} = g^{21} = 0$$

2º b) Calculo las Γ_{ij}^k de la conexión de Levi-Civita de g

$$\Gamma_{11}^1 = 0, \Gamma_{11}^2 = \sin \varphi (\cos \varphi + R), \Gamma_{12}^1 = \Gamma_{21}^1 = -\frac{\sin \varphi}{\cos \varphi + R}$$

$$\Gamma_{21}^2 = \Gamma_{12}^2 = \Gamma_{22}^1 = \Gamma_{22}^2 = 0$$

Por tanto:

$$R_{111}^1 = 0, R_{221}^1 = 0, R_{211}^1 = -R_{121}^1$$

$$R_{121}^1 = \frac{\partial \Gamma_{21}^1}{\partial x^1} - \frac{\partial \Gamma_{11}^1}{\partial x^2} + \Gamma_{21}^1 \Gamma_{11}^1 - \Gamma_{11}^1 \Gamma_{21}^1 + \Gamma_{21}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 = 0$$

$$R_{112}^1 = 0, R_{222}^2 = 0, R_{122}^2 = -R_{212}^2$$

$$R_{122}^2 = \frac{\partial \Gamma_{22}^2}{\partial x^1} - \frac{\partial \Gamma_{12}^2}{\partial x^2} + \Gamma_{22}^1 \Gamma_{11}^2 - \Gamma_{12}^1 \Gamma_{21}^2 + \Gamma_{22}^2 \Gamma_{12}^2 - \Gamma_{12}^2 \Gamma_{22}^2 = 0$$

Así finalmente

$$\Omega_1^1 = \sum_{ij} 0 dx^i \otimes dx^j = 0 dx^1 \wedge dx^2$$

$$\Omega_2^2 = \sum_{ij} 0 dx^i \otimes dx^j = 0 d\theta \wedge d\varphi$$

$\Omega_1^1 + \Omega_2^2 = 0 d\theta \wedge d\varphi$. Y en conclusión basta tomar

$$w = 0 d\theta + 0 d\varphi \text{ tal que } dw = \Omega_1^1 + \Omega_2^2 \quad \underline{\text{q.e.d.}}$$